DEEP LEARNING PROJECT · DIV2K · EDSR

IMAGE SUPER-RESOLUTION ×4

From a low-resolution image to a sharp ×4 reconstruction, learned from data.

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+2.32 dB
Y-PSNR over bicubic · 30.41 dB

−32%
LPIPS ↓ · 0.2804 vs 0.4139

EDSR PIPELINE

Low-resolution input → **Deep residual feature extraction** → **Learned ×4 upsampling** → **HR ×4**

All feature extraction runs at low resolution; the upsampling filters are learned from data, not fixed in advance.

DIV2K ×4 · 100 held-out test images

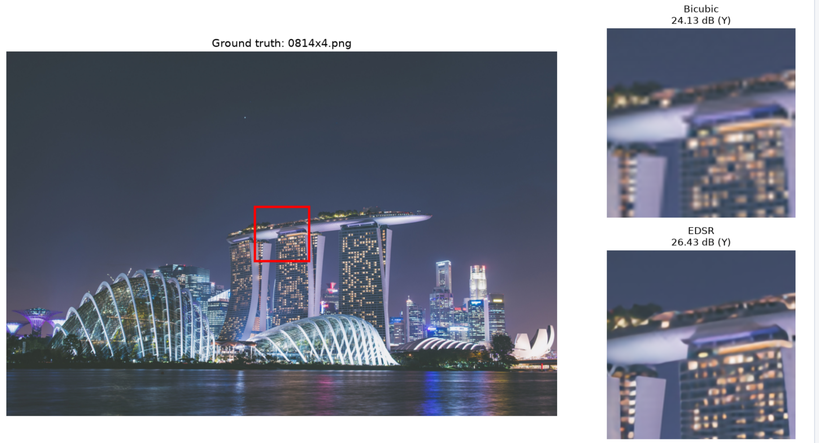
	Y-PSNR	Y-SSIM	LPIPS ↓
Bicubic	28.09	0.7744	0.4139
SRCNN	27.15	0.7780	–
VDSR	28.00	0.7960	–
EDSR	**30.41**	**0.8371**	**0.2804**
+ 8× self-ensemble	**30.48**	**0.8384**	0.2828

FIXED FOR THE SECOND SESSION

Clean train / val / test split · correct MATLAB Y-channel (+1.32 dB scale) · one evaluation path · MATLAB-matched bicubic baseline · fully seeded & reproducible · self-ensemble actually runs

Same model, same recipe — only the measurement changed.

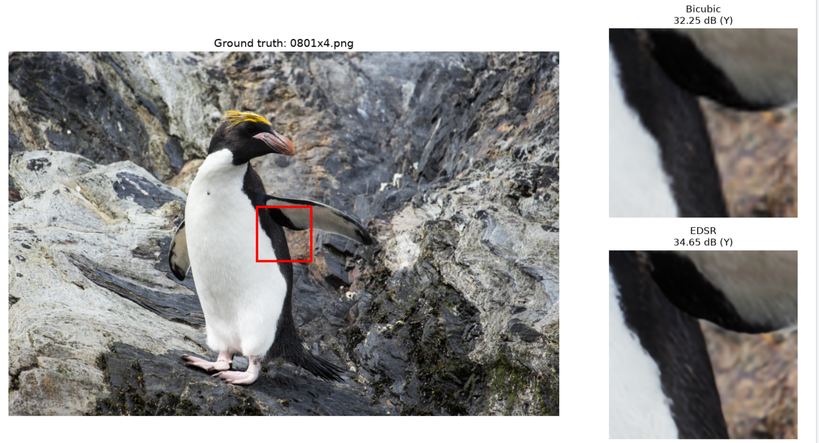
BICUBIC vs EDSR — DIV2K test crops



0814 · geometric structures · +2.30 dB



0843 · fine textures · +5.31 dB



0801 · edges and fur · +2.40 dB